# Ch 13 Gamma Functions

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## 13.1 Definition & Properties

integer $n$

$\Gamma(n) = (n-1)!$

### Infinite limit (Euler)

$$\Gamma(z) = \lim_{n\to\infty} \frac{1\cdot 2\cdot 3\cdots n}{z(z+1)\cdots(z+n)}\, n^z \qquad z \neq 0, -1, -2, \cdots$$

$$\Gamma(z+1) = \lim_{n\to\infty} \frac{1\cdot 2\cdot 3\cdots n}{(z+1)(z+2)\cdots(z+n)(z+n+1)}\, n^{z+1}$$

$$= \lim_{n\to\infty} \frac{nz}{z+n+1} \lim_{n\to\infty} \frac{1\cdot 2\cdot 3\cdots n}{z(z+1)\cdots(z+n)}\, n^z = z\,\Gamma(z)$$

$$\Gamma(1) = \lim_{n\to\infty} \frac{1\cdot 2\cdot 3\cdots n}{1\cdot 2\cdot 3\cdots n} \frac{n}{n+1} = 1 \Rightarrow \Gamma(2) = 1\cdot 1 \Rightarrow \Gamma(3) = 2\cdot 1$$

$$\Gamma(4) = 3\cdot 2\cdot 1 \Rightarrow \Gamma(5) = 4!$$

# Infinite Product (Weierstrass)

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$$\frac{1}{\Gamma(z)} \equiv z\, e^{\gamma z} \prod_{n=1}^{\infty} \left(1+\frac{z}{n}\right) e^{-z/n}$$

$\gamma \equiv$ Euler – Mascheroni Const. $\sim 0.5772156649\ldots$

$$\Gamma(z) = \lim_{n\to\infty} \frac{1}{z} \frac{1\cdot 2\cdot 3 \cdots n}{(z+1)(z+2)\cdots(z+n)} n^z = \lim_{n\to\infty} \frac{1}{z} \prod_{m=1}^{n} \left(1+\frac{z}{m}\right)^{-1} n^z$$

$$\frac{1}{\Gamma(z)} = z \lim_{n\to\infty} n^{-z} \prod_{m=1}^{n} \left(1+\frac{z}{m}\right)$$

Multi by $\prod_{m=1}^{n} e^{z/m} = \exp\left(\sum_{m=1}^{n} \frac{z}{m}\right)$ & Divide

# Weierstrass

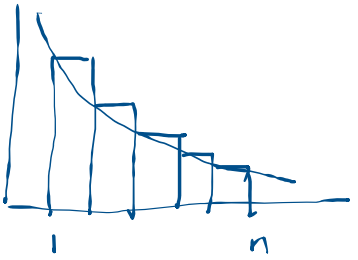
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$$\frac{1}{\Gamma(z)} = z \lim_{n\to\infty} e^{-z/n} \sum_{m=1}^{n} \frac{z}{m} \prod_{m=1}^{n} \left(1+\frac{z}{m}\right) e^{-z/m}$$

$$\frac{1}{\Gamma(z)} = z\, e^{\gamma z} \prod_{m=1}^{\infty} \left(1+\frac{z}{m}\right) e^{-z/m}$$

$$e^{z\left(\sum_{m=1}^{n} \frac{1}{m} - \ln n\right)}$$

$$\lim_{n\to\infty} \left(\sum_{m=1}^{n} \frac{1}{m} - \ln n\right) = \gamma$$



# Definite Int. (Euler)

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$$\Gamma(z) = \int_0^\infty e^{-t}\, t^{z-1}\, dt, \qquad \mathrm{Re}(z) > 0$$

$$\Gamma(z) = 2\int_0^\infty e^{-t^2}\, t^{2z-1}\, dt \rightarrow z = \tfrac{1}{2} \Rightarrow \text{Gaussian}$$

$$\Gamma(z) = \int_0^1 \left[\ln\left(\tfrac{1}{t}\right)\right]^{z-1} dt$$

$$\Gamma\left(\tfrac{1}{2}\right) = \sqrt{\pi}$$

$$\Gamma\left(\tfrac{3}{2}\right) = \tfrac{1}{2}\sqrt{\pi} \qquad \Gamma\left(\tfrac{5}{2}\right) = \tfrac{3}{2}\,\tfrac{1}{2}\sqrt{\pi}$$

## Connection

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$$F(z,n) = \int_0^n \left(1 - \frac{t}{n}\right)^n t^{z-1}\, dt$$

$$\lim_{n\to\infty} F(z,n) = F(z,\infty) = \int_0^\infty e^{-t} t^{z-1}\, dt = \Gamma(z)$$

Sub. $u = \frac{t}{n}$

$$F(z,n) = n^z \int_0^1 (1-u)^n u^{z-1}\, du$$

Integ. by parts:

$$\frac{F(z,n)}{n^z} = (1-u)^n \frac{u^z}{z}\Big|_0^1 + \frac{n}{z}\int_0^1 (1-u)^{n-1} u^z\, du$$

$$= \frac{n(n-1)\cdots 1}{z(z+1)\cdots(z+n-1)} \int_0^1 u^{z+n-1}\, du$$

$$F(z,n) = \frac{n(n-1)\cdots 1}{z(z+1)\cdots(z+n)} n^z \Rightarrow \lim_{n\to\infty} F(z,n) = \Gamma(z)$$

# Properties

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$$\Gamma(z+1) = z\,\Gamma(z) \quad \Rightarrow \quad n! = n(n-1)!$$

## Reflection formula

$$\Gamma(z)\,\Gamma(1-z) = \frac{\pi}{\sin \pi z}$$

**Proof**

$$\Gamma(z+1)\,\Gamma(1-z) = \int_0^\infty s^z e^{-s}\,ds \int_0^\infty t^{-z} e^{-t}\,dt$$

$$= \iint_0^\infty \left(\frac{s}{t}\right)^z e^{-(s+t)}\,ds\,dt$$

$u = s+t \qquad v = \frac{s}{t} \qquad v+1 = \frac{s+t}{t} = \frac{u}{t}$

$$ds\,dt = \left|\frac{\partial(s,t)}{\partial(u,v)}\right| du\,dv$$

$J \equiv$ Jacobian

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$v+1=\frac{u}{t}$ $\quad u=s+t,\ v=\frac{s}{t}$

$$|J^{-1}| = -\begin{vmatrix} \frac{\partial u}{\partial s} & \frac{\partial u}{\partial t} \\ \frac{\partial v}{\partial s} & \frac{\partial v}{\partial t} \end{vmatrix} = -\begin{vmatrix} 1 & 1 \\ \frac{1}{t} & -\frac{s}{t^2} \end{vmatrix} = \frac{s+t}{t^2} = \left(\frac{u}{t}\right)^2 \frac{1}{u}$$

$$= \frac{(v+1)^2}{u}$$

$$J = \frac{u}{(v+1)^2}$$

$$\Gamma(z+1)\Gamma(1-z) = \int_0^\infty \int_0^\infty v^z e^{-u} \frac{u}{(v+1)^2}\, du\, dv$$

$$= \int_0^\infty \frac{v^z}{(v+1)^2}\, dv \int_0^\infty u e^{-u}\, du$$

$\int_0^\infty u e^{-u}\, du = 1$

$\int_0^\infty \frac{v^z}{(v+1)^2}\, dv = \frac{\pi z}{\sin \pi z}$ ← Ex 11.8.20

$\Gamma(z+1)\Gamma(1-z) = \frac{\pi z}{\sin \pi z}$

$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$

Ex $\quad \Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right) = \frac{\pi}{\sin \frac{\pi}{2}} \Rightarrow \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

# Legendre Duplication Formula

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$$\Gamma(z+1)\,\Gamma\left(z+\tfrac{1}{2}\right) = 2^{-2z}\sqrt{\pi}\,\Gamma(2z+1)$$

## Proof for $z=n$ (integer)

$$\Gamma(n+1) = n! \qquad \Gamma(2n+1) = (2n)!$$

$$\Gamma\left(n+\tfrac{1}{2}\right) = \Gamma\left(\tfrac{1}{2}\right)\left[\tfrac{1}{2}\cdot\tfrac{3}{2}\cdot\tfrac{5}{2}\cdots\tfrac{2n-1}{2}\right]$$

$$= \sqrt{\pi}\,\frac{(1\cdot 3\cdot 5\cdots(2n-1))}{2^n}\,\frac{(2\cdot 4\cdot 6\cdots 2n)}{2^n(1\cdot 2\cdot 3\cdots n)}$$

$$= \sqrt{\pi}\,\frac{(2n)!}{2^{2n}\,n!}$$ ✓

Analytic Prop.

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**Weierstrass Def.** $\Gamma(z)$ has simple poles at $z = 0, -1, -2, \ldots$

$[\Gamma(z)]^{-1}$ has no poles $\Rightarrow$ $\Gamma(z)$ has no zeros

**Residues at $z = -n$ $(n \geq 0)$**

$$R_n = \frac{(-1)^n}{n!}$$

$z = -n + \varepsilon$

$$R_n = \lim_{z \to -n} (z+n)\,\Gamma(z)$$

$$= \lim_{\varepsilon \to 0} \varepsilon\,\Gamma(-n+\varepsilon) = \lim_{\varepsilon \to 0} \varepsilon \frac{\Gamma(-n+\varepsilon+1)}{-n+\varepsilon} = \frac{(-1)^n}{n!}$$

$$= \lim_{\varepsilon \to 0} \varepsilon \frac{\Gamma(-n+\varepsilon+2)}{(-n+\varepsilon)(-n+1+\varepsilon)} = \lim_{\varepsilon \to 0} \varepsilon \frac{\Gamma(1+\varepsilon)}{(-n+\varepsilon)\cdots(-1+\varepsilon)\,\varepsilon} = \frac{1}{(-1)^n\, n(n-1)\cdots 1}$$

# Schlaefli Integral

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$$\int_C e^{-t}\, t^{\nu}\, dt = \left(e^{2\pi i \nu} - 1\right) \Gamma(\nu+1)$$

for non-integer $\nu$

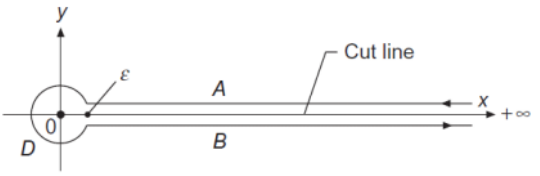



FIGURE 13.2 Gamma function contour.

Analytic continuation of Euler's integral

$(\nu > -1)$

Factorial Notation $\Gamma(z+1) = z!$

Gauss notation $\Pi(z) = z!$

Ex 13.1.1

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# MB Dist.

$$\mathcal{P}(E) = C e^{-\beta E}$$

# states $(E \to E+dE) = n(E)\,dE$

Prob. states ( " ) $= C\, n(E)\, e^{-\beta E}\, dE$

Normalization $1 = C \int n(E)\, e^{-\beta E}\, dE$

Avg. E $\langle E \rangle = C \int E\, n(E)\, e^{-\beta E}\, dE$

Let $n(E) \sim E^{1/2}$ ( Ideal gas )

$$1 = C \int_0^\infty E^{1/2} e^{-\beta E}\, dE$$

$$= C\, \frac{\Gamma(3/2)}{\beta^{3/2}} = C\, \frac{\sqrt{\pi}}{2\beta^{3/2}}$$

$$\Rightarrow C = \frac{2\beta^{3/2}}{\sqrt{\pi}}$$

$$\langle E \rangle = C \int_0^\infty E^{3/2} e^{-\beta E}\, dE$$

$$= C\, \frac{\Gamma(5/2)}{\beta^{5/2}}$$

$$= \frac{2\beta^{3/2}}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}\,(\frac{3}{2}\cdot\frac{1}{2})}{\beta^{5/2}} = \frac{3}{2}\frac{1}{\beta}$$

$$= \frac{3}{2} kT$$

# 13.2 Digamma & Polygamma function

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$$\Gamma(z+1) = z\,\Gamma(z) = \lim_{n\to\infty} \frac{n!}{(z+1)(z+2)\cdots(z+n)}\, n^z$$

$$\ln\Gamma(z+1) = \lim_{n\to\infty} \left[\ln n! + z\ln n - \ln(z+1) - \ln(z+2) - \cdots + \ln(z+n)\right]$$

$$\frac{d}{dz}\ln\Gamma(z+1) \equiv \underline{\psi(z+1)} = \lim_{n\to\infty}\left[\ln n - \frac{1}{z+1} - \frac{1}{z+2} + \cdots + \frac{1}{z+n}\right]$$

digamma function

$$H_n = \sum_{m=1}^{n} \frac{1}{m}$$

$$\psi(z+1) = \frac{d}{dz}\ln(\Gamma(z+1)) = \frac{\Gamma'(z+1)}{\Gamma(z+1)}$$

# Digamma

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$$\psi(z+1) = \lim_{n\to\infty}\left[\ln n - H_n\right] - \sum_{m=1}^{\infty}\left(\frac{1}{z+m} - \frac{1}{m}\right)$$

$$= -\gamma + \sum_{m=1}^{\infty}\frac{z}{m(m+z)}$$

$$\zeta(n) = \sum_{m=1}^{\infty}\frac{1}{m^n}$$

$$\boxed{\psi(1) = -\gamma}$$

$$\psi(n+1) = -\gamma + H_n = -\gamma + \sum_{m=1}^{n}\frac{1}{m}$$

For integer $z = n$

Polygamma Function

$$\psi^{(m)}(z+1) = \frac{d^{m+1}}{dz^{m+1}}\ln\Gamma(z+1) = (-1)^{m+1}\sum_{n=1}^{\infty}\frac{1}{(z+n)^{m+1}}$$

Maclaurin expansion

$$\operatorname{Ln}\Gamma(z+1) = \sum_{n=1}^{\infty}\frac{z^n}{n!}\psi^{(n-1)}(1) = -\gamma z + \sum_{n=2}^{\infty}(-1)^n\frac{z^n}{n!}\zeta(n)$$

# Beta function

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$$B(p,q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)} = 2\int_0^{\pi/2} \cos^{2p-1}\theta \sin^{2q-1}\theta \, d\theta$$

$$\Gamma(p)\Gamma(q) = 4\int_0^{\infty} s^{2p-1} e^{-s^2} ds \int_0^{\infty} t^{2q-1} e^{-t^2} dt$$

"Polar coord." $s = r\cos\theta \quad t = r\sin\theta \Rightarrow ds\,dt = r\,dr\,d\theta$

$$= 4\int_0^{\infty} r^{2p+2q-1} e^{-r^2} dr \int_0^{\pi/2} \cos^{2p-1}\theta \sin^{2q-1}\theta \, d\theta$$

$$= 2\Gamma(p+q)\int_0^{\pi/2} \cos^{2p-1}\theta \sin^{2q-1}\theta \, d\theta$$

$$B(p,q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)} = 2\int_0^{\pi/2} \cos^{2p-1}\theta \sin^{2q-1}\theta \, d\theta$$

For integers $p = m+1$, $q = n+1$

$$\frac{m!\,n!}{(m+n+1)!} = 2\int_0^{\pi/2} \cos^{2m+1}\theta \sin^{2n+1}\theta \, d\theta$$

"odd powers"

For half integers $p = \frac{m+1}{2}$, $q = \frac{n+1}{2}$

$$\frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{m+n}{2}+1\right)} = 2\int_0^{\pi/2} \cos^m\theta \sin^n\theta \, d\theta$$

"even powers"

Symmetry $B(p,q) = B(q,p)$

# Beta function

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## Alternative Definite integrals

$t = \cos^2\theta \Rightarrow B(p+1, q+1) = \int_0^1 t^p (1-t)^q \, dt$

$t = x^2 \Rightarrow B(p+1, q+1) = 2\int_0^1 x^{2p+1} (1-x^2)^q \, dx$

$t = \frac{u}{1+u} \Rightarrow B(p+1, q+1) = \int_0^\infty \frac{u^p}{(1+u)^{p+q+2}} \, du$

# Legendre Duplication Formula

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$$B(z+\tfrac{1}{2}, z+\tfrac{1}{2}) = \int_0^1 t^{z-\frac{1}{2}} (1-t)^{z-\frac{1}{2}} dt$$

$t = \frac{1+s}{2} \Rightarrow dt = ds/2 \qquad 1-t = \frac{1-s}{2}$

$$B(z+\tfrac{1}{2}, z+\tfrac{1}{2}) = 2^{-2z} \int_{-1}^{1} (1-s^2)^{z-\frac{1}{2}} ds = 2^{-2z} \, 2 \int_0^1 (1-s^2)^{z-1/2} ds$$

$$B(z+\tfrac{1}{2}, z+\tfrac{1}{2}) = 2^{-2z} B(\tfrac{1}{2}, z+\tfrac{1}{2})$$

$$\frac{\Gamma(z+\frac{1}{2})\,\Gamma(z+\frac{1}{2})}{\Gamma(2z+1)} = 2^{-2z} \frac{\Gamma(\frac{1}{2})\,\Gamma(z+\frac{1}{2})}{\Gamma(z+1)}$$

$$\boxed{\Gamma(z+1)\,\Gamma(z+\tfrac{1}{2}) = \frac{\sqrt{\pi}}{2^{2z}}\,\Gamma(2z+1)}$$